KEY FACTS

TERMINOLOGY

- Lysosomal storage disease resulting in dense bones

IMAGING

- Dense bones with short stature, acroosteolysis, decreased angle of mandible
  - Osteosclerosis is dominant radiographic finding
- Skull
  - Wormian bones
  - Thick skull base
  - Thin cranial vault
  - Delayed closure of calvarial sutures
  - Persistent anterior fontanelle
- Poorly developed midface
  - Obtuse mandibular angle
- Hypoplastic/malformed clavicles
- Fragility fractures
  - Spondylolysis
- Acroosteolysis terminal phalanges

TOP DIFFERENTIAL DIAGNOSES

- Osteopetrosis
  - Primary differential diagnostic consideration
  - Bone-in-bone appearance
  - Obliterates marrow space
  - No acroosteolysis or mandibular deformity
- Progressive diaphyseal dysplasia
  - Cortical thickening
  - Bilateral symmetric distribution
  - Typically confined to diaphyses

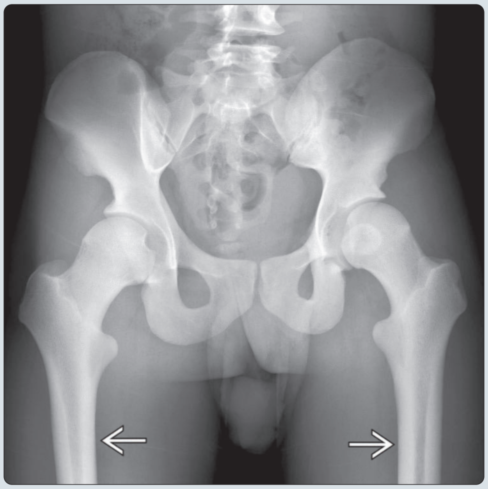
PATHOLOGY

- Mutation in cathepsin K enzyme
- Autosomal recessive

CLINICAL ISSUES

- Osteomyelitis of mandible; abnormal tooth development
- Midface hypoplasia with associated sleep apnea
- Transverse fragility fractures of long bones

(Left) *Lateral radiograph shows an abnormal mandibular angle ➡. The normal angle is 110° to 120°. In pyknodysostosis, the angle becomes more obtuse, approaching 180°. This mandibular anomaly leads to complications, such as sleep apnea.* **(Right)** *AP radiograph shows severe thickening of the endosteum of the femora ➡. In isolation, this finding is nonspecific. When associated with resorption of the terminal tufts of the phalanges (not shown), the radiographs are diagnostic of pycnodysostosis.*



(Left) *AP radiograph reveals diffuse sclerosis of the radius and ulna accompanied by diffuse endosteal thickening ➡. Mild undertubulation of the distal radius is present ➡. Undertubulation is more pronounced and prevalent in osteopetrosis and is an inconsistent finding in pycnodysostosis.* **(Right)** *PA x-ray shows diffuse sclerosis and significant resorption of the terminal tufts of all of the distal phalanges ➡. In this case, the osteolysis is severe and uniformly distributed. Acroosteolysis is typical in pycnodysostosis.*

TERMINOLOGY

Synonyms

- Maroteaux-Lamy

Definitions

- Lysosomal storage disease resulting in dense bones

IMAGING

General Features

- Best diagnostic clue
  - Dense bones with short stature, acroosteolysis, decreased angle of mandible
- Location
  - Involves entire skeleton

Imaging Recommendations

- Best imaging tool
  - Radiographs are diagnostic

Radiographic Findings

- Osteosclerosis is dominant radiographic finding
  - Does not obliterate marrow space
- Acroosteolysis
  - Resorption or band-like lucency in terminal tuft
  - Asymmetric distribution
  - May not involve all digits
  - May be progressive
- Skull
  - Wormian bones
  - Thick skull base
  - Thin cranial vault
  - Delayed closure of calvarial sutures
    - Persistent anterior fontanelle
- Face
  - Poorly developed midface
  - Hypoplastic mandible
    - Obtuse mandibular angle
  - Hypoplastic paranasal sinuses
- Minimal undertubulation of long bones
- Hypoplastic/malformed clavicles, especially laterally
- Fragility fractures
  - Transverse in long bones
  - Poor healing
  - Spondylolysis

CT Findings

- Depict same changes identified on radiographs
- May be used to assess fracture healing
- Useful to detail anomalies, especially dental and facial

MR Findings

- May be useful to evaluate fracture healing/pseudoarthrosis

DIFFERENTIAL DIAGNOSIS

Osteopetrosis

- Primary differential diagnostic consideration
- Undertubulation may be marked, especially distal femur
- Bone-in-bone appearance
- Obliterates marrow space
  - Clinical symptoms of marrow failure

Progressive Diaphyseal Dysplasia

- Cortical thickening
  - Bilateral symmetric distribution
  - Typically confined to diaphyses
- Spares spine
- Thick calvaria

Ribbing Disease

- Endosteal sclerosis confined to lower extremities

PATHOLOGY

General Features

- Etiology
  - Mutation in cathepsin K enzyme
    - Enzyme required for bone turnover
    - Abnormal enzyme results in abnormal osteoclast activity, decreased bone resorption/turnover
    - Results in brittle bones, similar to osteopetrosis
- Genetics
  - Autosomal recessive

CLINICAL ISSUES

Presentation

- Most common signs/symptoms
  - Short stature
  - Hands are characteristic
    - Short and broad, irregular shortening secondary to acroosteolysis, wrinkled skin
- Other signs/symptoms
  - Large head
    - Frontal and occipital bossing
  - Hypoplastic nails
  - Fractures, often with minor trauma
    - Poor healing
  - Osteomyelitis of mandible
  - Abnormal tooth development
  - Midface hypoplasia with associated sleep apnea
  - Grooved palate
  - Prominent/beaked nose
  - No sequelae of marrow failure, in contrast to osteopetrosis
  - Probably condition of artist Toulouse-Lautrec

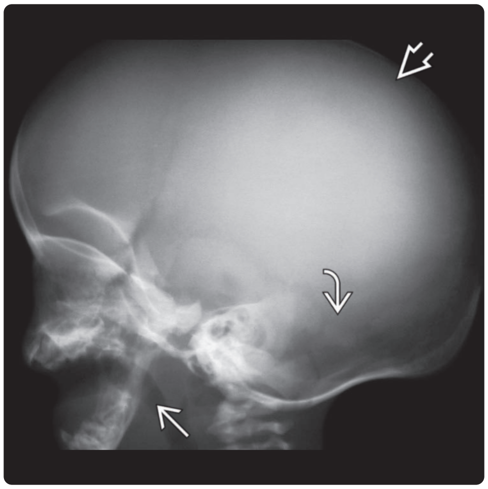
Demographics

- Epidemiology
  - Rare

SELECTED REFERENCES

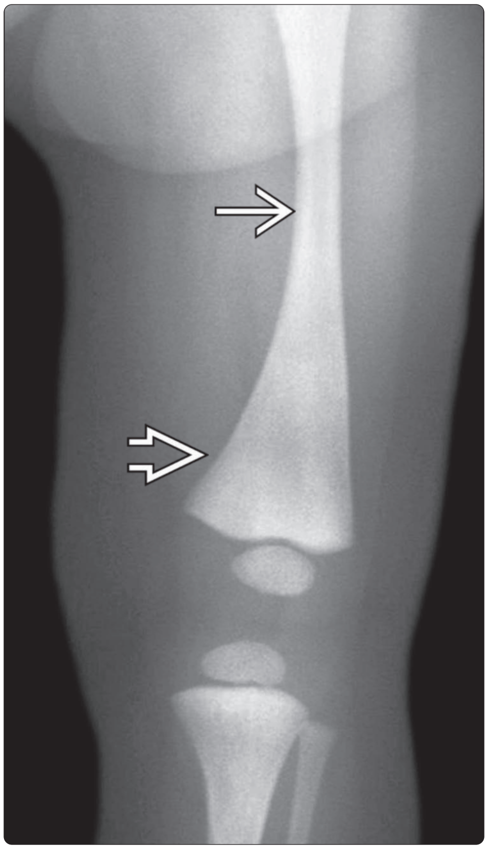
1. S R et al: Osteomyelitis in pycnodysostosis - report of 2 clinical cases. J Clin Diagn Res. 9(1):ZD15-7, 2015
2. Yates CJ et al: An atypical subtrochanteric femoral fracture from pycnodysostosis: a lesson from nature. J Bone Miner Res. 26(6):1377-9, 2011
3. Bathi RJ et al: Pyknodysostosis—a report of two cases with a brief review of the literature. Int J Oral Maxillofac Surg. 29(6):439-42, 2000
4. Vanhoenacker FM et al: Sclerosing bone dysplasias: genetic and radioclinical features. Eur Radiol. 10(9):1423-33, 2000

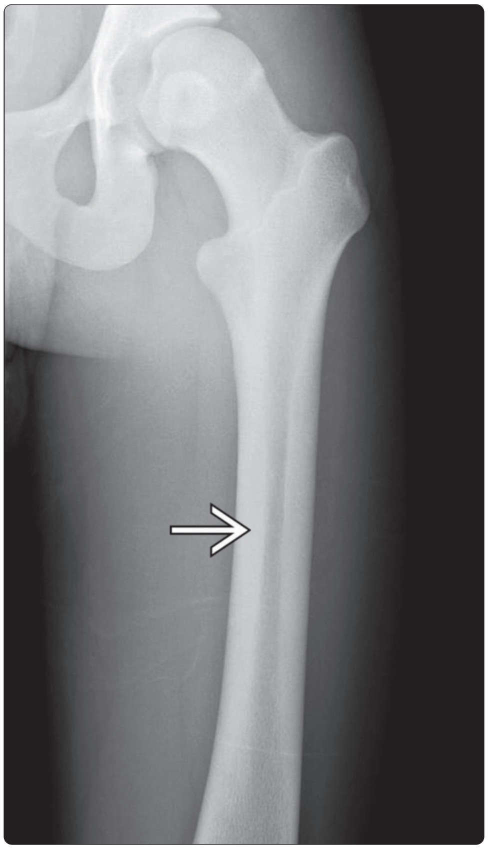
(Left) *Lateral radiograph demonstrates a shallow mandibular angle ➡, a characteristic finding in pyknodysostosis. Generalized thinning of the cranial vault ➡ can also be observed. Wormian bones are present although they are difficult to visualize on this image ➡.* **(Right)** *AP radiograph shows uniformly dense bone, which is a nonspecific finding. This may be seen in several sclerosing dysplasias, including osteopetrosis and pyknodysostosis. The latter was the diagnosis in this case.*



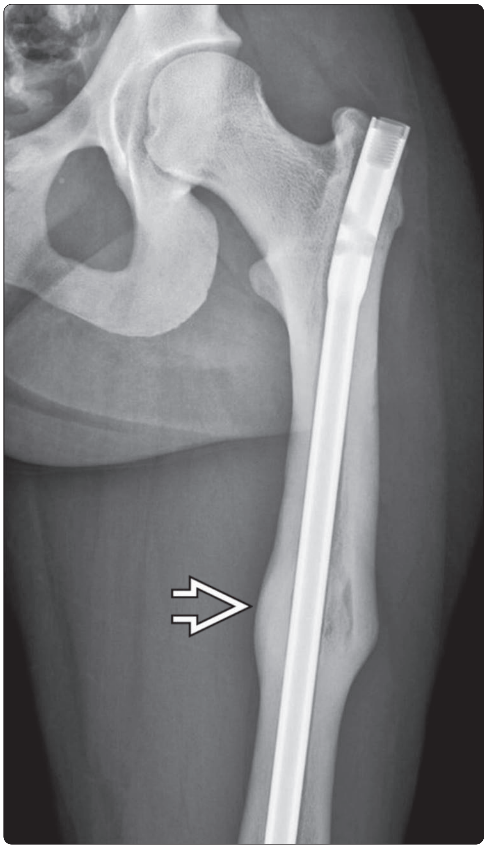
(Left) *PA radiograph shows diffusely dense bones associated with acroosteolysis ➡. In this young child, the resorption is minimal. Acroosteolysis may be a progressive process and can be more severe in adults.* **(Right)** *Lateral radiograph demonstrates acroosteolysis. Note that the resorption is not uniform, sparing the 3rd and 4th distal phalanges. Two patterns of resorption can be seen: Band-like resorption in the index finger ➡ and tuftal resorption of the 1st and 5th distal phalanges ➡.*

(Left) *AP radiograph is remarkable for diffusely dense bones, resorption of the terminal tufts of the distal phalanges ➡, and a fragility fracture at the base of the 5th metatarsal ➡. This combination of findings is diagnostic for pyknodysostosis.* **(Right)** *Sagittal bone CT reveals increased density throughout all of the vertebra. Multiple pars interarticularis defects are visible ➡. These fractures are one of the manifestations of bone fragility.*





(Left) *AP radiograph demonstrates diffusely dense bones of the lower extremity with poor cortical-medullary differentiation ➡. Mild undertubulation of the distal femur is present ➡. The appearance is nonspecific and leads to a diagnosis of sclerosing dysplasia. Pycnodysostosis and osteopetrosis are the leading diagnoses.* **(Right)** *AP radiograph nicely reveals diffuse endosteal thickening ➡, which is one of the manifestations of pyknodysostosis. The spectrum of dense bones in this disease ranges from endosteal thickening, as in this case, to diffusely dense bones, which mimic osteopetrosis. Evaluation of the hands and feet as well as the skull will help to differentiate these entities.*



(Left) *AP radiograph shows a femur with diffuse sclerosis and prominent endosteal thickening. Despite the thickened cortex, fracture has occurred (now healed) ➡. Even though the bone is dense, the disorganized architecture and the inability to properly remodel in response to normal stresses places these bones at ↑ risk for fracture.* **(Right)** *Lateral radiograph of the leg shows dense bone, as well as transverse fractures ➡. Transverse fractures either occur from significant trauma (not the case here), or in abnormal bone. Note how the fracture has occurred along the anterior tibial cortex at the junction of mid and distal 1/3, a site of tension stresses. This location is a common site for adult tibial stress fractures. Normal stress on dense but abnormally remodeled bone in this case of pycnodysostosis has resulted in pathologic fracture.*